

Exchange Certificates for an Inductive Approach to Triangle-Free Graph Bipartization

Jinji Li

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Abstract

For a graph G , let

$$d(G) = |E(G)| - \text{MaxCut}(G)$$

be the minimum number of edges whose deletion makes G bipartite. A classical conjecture of Erdős asserts that every triangle-free graph on N vertices satisfies $d(G) \leq N^2/25$; for $N = 5k$ this becomes $d(G) \leq k^2$.

We study an induction strategy based on deleting five vertices at a time, with target

$$\exists X \subseteq V(G), |X| = 5 : \quad d(G) - d(G - X) \leq 2k - 1.$$

The paper develops three complementary pieces of evidence for this route: exact balanced and mixed extension certificates, a coordinated-flip inequality that converts local recolorings into a selection criterion, and an exact 3-for-3 exchange audit in the difficult $n = 15$ regime.

A stronger fixed- X gain statement suggested by early experiments is false on explicit realizable residual configurations. The failure does not contradict the induction target: exact audits show that changing the five-set repairs the obstruction. In the saved $n = 15$ hard-regime corpus, all 3,436 canonical classes with global $L(G) = 11$ admit, relative to the recorded distinguished five-set X , a five-set Y with $|X \cap Y| = 2$ and

$$d(G) - d(G - Y) \leq 5.$$

Among these classes, 762 have $d(G) > 5$ and therefore require a nontrivial search; all 762 pass.

The full Erdős conjecture remains open. The results here are structural and computational certificates for one induction mechanism, not a proof of the conjecture.

1 Introduction

For a finite simple graph G , define its edge-bipartization number by

$$d(G) = \min\{|F| : F \subseteq E(G), G - F \text{ is bipartite}\} = |E(G)| - \text{MaxCut}(G).$$

Erdős conjectured that every triangle-free graph on N vertices satisfies

$$d(G) \leq \frac{N^2}{25}.$$

The balanced blow-up of C_5 shows that the constant $1/25$ would be sharp. Balogh, Clemen and Lidický proved the global bound $N^2/23.5$ and proved the sharp $N^2/25$ bound in the two density tails [1]. A recent computer-assisted preprint of Ferudun proves the sharp value for every multiple

of five up to $N = 200$ [2]. Consequently, the small finite cases used below are validation and structure-finding devices, not new extremal evaluations.

The present project asks a different question: can the sharp bound be approached by a five-vertex induction that mirrors the extremal C_5 blow-up? For a triangle-free graph on $5k$ vertices, the natural target is as follows.

Conjecture 1 (Candidate A). *For every triangle-free graph G on $5k$ vertices, there exists a five-set $X \subseteq V(G)$ such that*

$$d(G) - d(G - X) \leq 2k - 1.$$

If Candidate A holds for all k , induction from $k = 1$ gives

$$d(G) \leq 1 + 3 + \cdots + (2k - 1) = k^2.$$

Thus the problem becomes one of *selecting* a good five-set, not only of bounding a fixed extension.

2 Exact finite ground truth

All exact computations use the identity

$$d(G) = |E(G)| - \text{MaxCut}(G).$$

For a graph on ten vertices, global color-complement symmetry allows one vertex to be fixed, so a direct MaxCut check uses at most 2^9 colorings. As a ground-truth computation, the 12,172 triangle-free isomorphism classes on ten vertices were checked exactly; the maximum is

$$\max d(G) = 4,$$

attained by the balanced C_5 blow-up B_2 .

This computation is useful internally because it supplies an exact base case and a regression test for later routines. In view of [2], it is not presented as a novelty claim.

3 Balanced and mixed extension certificates

The first reduction studies extensions organized around the five-part structure suggested by balanced C_5 blow-ups. The implementation separates a balanced extension model from mixed attachment patterns. Because these are finite reduced state spaces, the conclusions in this section are exact computational certificates for the represented models.

3.1 Balanced-extension certificate

For each parameter value $t = 1, 2, \dots, 25$, the exact C++ checker `scripts/check_balanced_extension.cpp` enumerates the reduced balanced-extension states and compares the exact deletion distance with the target inequality. In every batch the reported maximum gap is zero:

$$\text{max_gap} = 0,$$

and the checker reports that the inequality holds for every represented state.

Proposition 1 (Balanced-extension certificate). *For every balanced-extension state represented by the reduction at $t = 1, \dots, 25$, the target inequality checked by the exact program holds. No violating state occurs in the enumerated reduced state space.*

The significance of this certificate is methodological rather than a stand-alone theorem about arbitrary graphs: it shows that the balanced part of the proposed induction behaves exactly as the C_5 blow-up heuristic predicts throughout the tested range.

3.2 Exact mixed $t = 2$ closure

The mixed $t = 2$ model is the first regime in which different attachment types interact. The exact closure contains

$$4,013,521$$

maximal-pair states, of which

$$2,768,154$$

are genuinely mixed. Exact MaxCut evaluation gives maximum deletion distance 9 over the full state space and maximum deletion distance 8 among the genuinely mixed states.

A separate maximal-completion monotonicity argument is used in the code: once a legal mixed state is extended to a maximal triangle-free completion, the completed state is the only one that must be checked for the relevant upper-bound test. Thus the exhaustive maximal-state computation closes the arbitrary legal mixed $t = 2$ states represented by this reduction.

Proposition 2 (Mixed $t = 2$ certificate). *Within the exact mixed $t = 2$ extension model, all legal states are closed by maximal triangle-free completion. Among the maximal states the largest deletion distance is 9, while the largest genuinely mixed value is 8. In particular, no mixed $t = 2$ obstruction to the tested induction inequality occurs in this model.*

These finite certificates motivated the next step: instead of forcing a single extension coloring, allow the core coloring to change in a controlled way.

4 Coordinated flips and the selection inequality

Fix a five-set $X \subseteq V(G)$ and write $H = G - X$. Let c be an optimal 2-coloring of H , so that the number of monochromatic edges of H under c is $d(H)$. Let a be a 2-coloring of X . Denote by $E_c(a)$ the number of monochromatic edges contributed by $G[X]$ and the cut $E(X, H)$ before modifying the core coloring.

For $v \in V(H)$ define

$$w_a(v) = \#\{\text{monochromatic edges from } v \text{ to } X\} - \#\{\text{bichromatic edges from } v \text{ to } X\},$$

and

$$\sigma_c(v) = \#\{\text{bichromatic core edges incident with } v\} - \#\{\text{monochromatic core edges incident with } v\}.$$

For $S \subseteq V(H)$ let $m_c(S)$ be the number of monochromatic edges of $H[S]$ under c . If all colors on S are flipped, direct edge accounting gives the exact gain

$$g_c(a, S) = \sum_{v \in S} (w_a(v) - \sigma_c(v)) + 2e(H[S]) - 4m_c(S).$$

Therefore

$$b_G(c^S \cup a) - d(H) = E_c(a) - g_c(a, S), \tag{1}$$

where $b_G(\cdot)$ denotes the number of monochromatic edges of G under the displayed coloring.

The implemented admissible flip family consists of independent sets and sets inducing complete bipartite subgraphs, together with their complements. This is deliberately narrower than the family of all bipartite induced sets; an earlier audit that used the larger family was corrected before the results reported here.

Let μ_X be the uniform distribution on optimal 2-colorings of $G[X]$, counting color reversals separately. For every fixed optimal core coloring c ,

$$\mathbb{E}_{a \sim \mu_X} E_c(a) = d(G[X]) + \frac{e(X, H)}{2}. \quad (2)$$

Define

$$\Phi_X = \max_{c \text{ optimal on } H} \mathbb{E}_{a \sim \mu_X} \max_{S \text{ admissible}} g_c(a, S).$$

Combining (1) and (2) gives the selection bound.

Proposition 3 (Coordinated-flip selection bound). *For every five-set X ,*

$$d(G) - d(G - X) \leq \left\lfloor d(G[X]) + \frac{e(X, G - X)}{2} - \Phi_X \right\rfloor. \quad (3)$$

Consequently, for a graph on $5k$ vertices, Candidate A holds for X if

$$\Phi_X \geq d(G[X]) + \frac{e(X, G - X)}{2} - (2k - 1).$$

Proof. For every optimal core coloring c and every assignment a on X , choose an admissible flip maximizing $g_c(a, S)$. The resulting coloring of G has cost at most the right-hand side of (1). Since $d(G)$ is the minimum cost over all 2-colorings, it is at most the average of these constructed costs over $a \sim \mu_X$. Equation (2) gives the displayed bound, and integrality gives the floor. Finally maximize over the choice of optimal core coloring c . $\square$

For $k = 3$, write

$$\ell(X) = 2d(G[X]) + e(X, G - X), \quad L(G) = \min_{|X|=5} \ell(X).$$

If $L(G) \leq 10$, the empty flip already yields the desired bound $d(G) - d(G - X) \leq 5$. Thus the genuinely difficult $n = 15$ regime begins at $L(G) > 10$.

5 The saved $n = 15$ hard regime

The saved canonical corpus contains 8,773 graphs with $L(G) > 10$. Their L -distribution is recorded by the exact audit; in particular, exactly

3,436

classes have global $L(G) = 11$.

For the $L = 11$ exchange analysis, the stored distinguished five-set need not itself minimize ℓ . In the 3,436 records, the distinguished sets have

$$\ell(X) \in \{11, 12, 13, 14, 15\}$$

with counts

2502, 636, 244, 48, 6,

respectively. This distinction is important: “global $L(G) = 11$ ” means that some five-set realizes 11, not that every later distinguished set has $\ell = 11$.

The scope of the exchange file was independently checked against the global L records. Both contain exactly 3,436 global- $L = 11$ graph IDs; the ID sets agree, there are no missing or extra records, and the canonical graph6 strings agree record by record.

6 Why a stronger fixed-set lemma fails

The first $L = 11$ audits suggested a stronger statement than Candidate A. For a distinguished C_5 five-set, there are ten optimal assignments when color reversals are counted. Empirically, the initial saved-corpus audit had minimum average gain 0.7, suggesting a total gain of at least 7 over those ten assignments.

That stronger claim is false. A direct actual-core analysis reduces the remaining low-imbalance configurations to 22 signed-boundary orbits. Exact realization and minimization show that all 22 admit realizable configurations with total gain below 7. Thus the proposed universal “total gain at least 7” lemma cannot be used.

The correction is important because the stronger lemma is not the actual selection threshold. For a C_5 set X in the $k = 3$ case, (3) requires

$$\Phi_X \geq \frac{\ell(X) - 10}{2}.$$

When $\ell(X) = 11$, the required average gain is only $1/2$, equivalently a total gain of 5 over the ten optimal assignments. Several residual orbits attain this smaller threshold exactly. Therefore the 7-gain lemma is false while Candidate A remains intact.

The 22 realizable residual witnesses were then checked without fixing the original five-set. An exact audit over all $\binom{15}{5} = 3003$ deletion sets per witness gives

$$\max q_{\min} = 4, \quad q(Y) = d(G) - d(G - Y),$$

so none is a Candidate-A obstruction. More strikingly, restricting to 3-for-3 exchanges relative to the original five-set gives the same minimum distribution across the 22 witnesses:

restricted minimum q	1	2	3	4
number of witnesses	1	10	8	3

Every residual witness therefore has a global minimizer obtainable by retaining exactly two vertices of the original X and replacing the other three. This observation motivated the full-corpus exchange test.

7 The 3-for-3 exchange certificate

A *3-for-3 exchange* of a recorded five-set X is a five-set $Y \subseteq V(G)$ satisfying

$$|X \cap Y| = 2.$$

There are

$$\binom{5}{2} \binom{10}{3} = 1200$$

such sets when $|V(G)| = 15$.

Proposition 4 (Exact global- $L = 11$ corpus certificate). *For every one of the 3,436 canonical graph classes in the saved $n = 15$ hard-regime corpus with global $L(G) = 11$, the recorded distinguished five-set X admits a 3-for-3 exchange Y such that*

$$d(G) - d(G - Y) \leq 5.$$

The complete deletion-distance distribution of the 3,436 classes is

$d(G)$	0	1	2	3	4	5	6	7
#	5	49	160	472	856	1132	701	61

For the 2,674 classes with $d(G) \leq 5$, the conclusion is automatic because $d(G - Y) \geq 0$. The remaining

$$762 = 701 + 61$$

classes are nontrivial. The exact C++ checker searches 3-for-3 sets in a deterministic order and passes all 762.

For these nontrivial classes, the first successful exchange encountered by the early-exit search has

$q(Y)$	3	4	5
#	26	307	429

These numbers are *first-success values*, not restricted minima; no minimality claim is attached to this distribution.

Remark 1. *The proposition is exhaustive for the 3,436 global- $L = 11$ records of the saved hard-regime corpus. It does not, by itself, certify that the larger saved $n = 15$ corpus exhausts all triangle-free isomorphism classes on 15 vertices. The scope statement is intentionally kept separate from any stronger universe-completeness claim.*

The computational result suggests the following structural target.

Conjecture 2 (3-for-3 exchange principle). *Under an appropriate structural hypothesis capturing the global- $L = 11$ regime, a distinguished five-set X admits a five-set Y with $|X \cap Y| = 2$ and*

$$d(G) - d(G - Y) \leq 5.$$

A proof of such a statement, combined with a reduction of a minimal Candidate-A counterexample into the relevant structural regime, would turn the present finite certificate into a genuine induction lemma.

8 Reproducibility and certificate scope

The current computation is organized so that the numerical claims above can be regenerated from explicit finite inputs. The principal artifacts are:

- `results/hard_regime_L_n15/L_records.tsv`, containing the global L records;
- `results/hard_regime_L_n15/L11_exchange.jsonl`, containing the 3,436 exchange-analysis records;
- `scripts/build_L11_3for3_input.py`, which constructs the deterministic C++ input;

- `scripts/check_L11_3for3_all.cpp`, the exact 3-for-3 checker;
- `results/hard_regime_L_n15/L11_3for3_input.txt` and `L11_3for3_full_audit.tsv`, the generated input and audit output.

The release checker independently recomputes $d(G)$ using exact MaxCut, verifies triangle-freeness, and then tests the exchange condition. The scope audit independently matches the global- $L = 11$ IDs and graph6 strings against the exchange file. In the release version these inputs and outputs should be accompanied by SHA-256 checksums and a one-command regression script.

The distinction between three levels of evidence is maintained throughout: (1) symbolic identities such as the flip formula, (2) exhaustive finite certificates for explicitly defined reduced state spaces or saved corpora, and (3) conjectural structural statements intended for future proof.

9 Discussion

The computations point to a useful change in perspective. A natural fixed-set strengthening can fail even when the existential five-set induction target is comfortably true. The residual 22-orbit analysis shows concretely how insisting on one distinguished set can create an artificial obstruction. The 3-for-3 audit then shows that a small exchange of the deletion set is sufficient across the entire saved global- $L = 11$ corpus.

Two proof directions appear particularly natural. First, an edge-minimal counterexample to the sharp bipartization bound would have highly constrained edge-critical behavior, potentially reducing the number of admissible local configurations. Second, the coordinated-flip formula separates signed boundary imbalance from exact internal flip corrections, providing a route to prove that one of a small family of exchanges must have enough gain.

The immediate goal is therefore not a broader brute-force search. It is to convert the observed exchange phenomenon into a lemma: identify a minimal set of structural hypotheses under which a 3-for-3 exchange must exist, and then derive those hypotheses from a minimal counterexample to Candidate A.

Status and non-claims

The Erdős triangle-free bipartization conjecture remains open in general. This manuscript does not claim to solve the conjecture or JSP-000058. The finite computations at $n = 10$ and $n = 15$ are not claimed as new extremal evaluations. The contribution is the exact certificate framework and the structural evidence for the five-set induction and 3-for-3 exchange mechanisms.

References

- [1] J. Balogh, F. C. Clemen, and B. Lidický, *Max Cuts in Triangle-Free Graphs*, arXiv:2103.14179, 2021.
- [2] A. Ferudun, *The Erdős $n^2/25$ max-cut conjecture for small multiples of five, via a per-root-MaxCut envelope and blow-up integrality*, arXiv:2606.28041, 2026.